

Belief-State Safety Values for Viability under Incomplete Observation

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Abstract

The obstruction calculus characterizes nonviability through certificates on information states; the probabilistic side asks what the certificates imply when the information state is a distribution. This paper develops the belief-state safety value $V_k(b)$ — the maximal probability of remaining safe for k steps under the disturbance’s worst case — and establishes four results. First, a support identity: on finite models with deterministic kernels the posterior support coincides with the set-valued post-state of the calculus, the value-one level of the recursion coincides with the viable-set recursion $\mathcal{W}_k$, and outside it the deficit satisfies $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$, attained on natural instances. Second, exactness: over rational input data the finite-horizon values are piecewise linear with rational alpha-vectors, so all values are exact rationals computable without floating point; the verification campaign’s 384 vectors are all exact, realizing exactly three types (156, 156, and 72 occurrences), and the complete 16×12 value table over the audited grid is carried here. Third, agreement on the audited delayed-regime class: a class-restricted alpha-vector recursion reproduces the closed-form values on all 48 audited cells and horizons $k \leq 4$, the certificate partition coincides with the value boundaries, and the value jumps by exactly $1/2$ at $z_0 = 1 + \min(k, T_{\text{obs}})$; the declared hold class equals the unrestricted sequential blind class at the one-step horizon, and for horizons $k \geq 2$ the two differ exactly on the timing cells ($2 \leq z_0 < 1 + T_{\text{obs}}$), where open-loop time-variation within the blind window lifts the value to one — 36 cell-horizon combinations in all, enumerated in full. Fourth, adaptive observation: on the declining hidden-parameter instance a single probe reveals the parameter exactly (branch separation 2), survival forces $z_0 \geq 21/10$, and under revelation at horizon T_{learn} viability holds exactly when $z_0 \geq 21/10 + T_{\text{learn}}/10$ — the learning deadline acts as the timing obstruction’s observation deadline, and the kernel shrinks linearly and antitone in it ($|K| = 5, 4, 3, 2, 1$). All computations run in exact rational arithmetic; this edition carries the complete audited tables, the derivations behind the closed form and the class declaration, and a five-figure set regenerated from the verification scripts.

Keywords: partially observable control; viability kernels; belief states; safety values; exact arithmetic; adaptive observation

1. Introduction

Viability under incomplete observation divides into a necessity side and a sufficiency side. The obstruction calculus treats the necessity side: certificates whose firing proves nonviability of an information state (Abaee, 2026b). The probabilistic question is the quantitative counterpart. If the controller holds a distribution b over states rather than a set, how safe is safe? The answer is a function: the belief-state safety value $V_k(b)$, the maximal worst-case probability of keeping the state trajectory in the safe set for k steps. Its value-one level is the viability statement of the set-valued theory; everything below level one is measured by the deficit $1 - V_k(b)$, and the obstruction certificates reappear as lower bounds on that deficit.

This paper develops the value function in five steps. Section 3. connects the belief-state recursion to the set-valued recursion of the calculus exactly, through the support of the belief. Section 4. shows the finite-horizon values are piecewise linear and exactly representable, and carries the complete audited value table over the delayed grid. Section 5. evaluates the deterministic limit, at which the recursion degenerates to a survived-mass formula over blind action sequences, with the asymmetric audit carried sequence by sequence. Section 6. reads the obstruction certificates systematically as deficit lower bounds. Section 7. proves agreement between the class-restricted value recursion and the closed-form values on the audited delayed-regime class, quantifies what the class declaration contributes, locates the value’s jump loci, and carries the derivations. Section 8. treats observation as a design variable: probes, learning deadlines, and the exact kernel cost of late resolution of a hidden parameter.

All audited computations are exact: integer and rational arithmetic, standard library only, deterministic scripts (Section 9.).

2. The belief-state model and the value recursion

The model is the calculus’s finite model with an absorbing unsafe state. Let X be a finite state set with a designated safe set $\mathcal{V} \subseteq X$, A a finite action set, Y a finite observation set, and D a finite disturbance set with deterministic transition and observation maps $x^+ = F(x, a, d)$ and $y = O(x, a, x^+)$; total transition rows over the augmented space $X \cup \{\partial\}$, with ∂ absorbing, unsafe, and car-

rying its own observation, convert set-valued nonviability into almost-sure nonviability on the augmented chain. A belief is a probability distribution b on X ; the posterior after action a and observation y is $b^+(x') \propto \sum_x b(x) \mathbb{K}[y = O(x, a, x')]$.

Definition 1 (safety value). For horizon k and belief b ,

$$V_k(b) = \sup_{\pi} \inf_{d\text{-strategies}} \mathbb{P}_b^{\pi}(x_t \in \mathcal{V} \text{ for all } t \leq k),$$

the supremum over observation-based policies and the infimum over disturbance strategies.

Theorem 2 (value recursion). $V_0(b) = \mathbb{K}[b \in \mathcal{P}(\mathcal{V})]$, and for $k \geq 1$,

$$V_k(b) = \sup_a \mathbb{E}_{y \sim b} V_{k-1}(b^+(\cdot | a, y)),$$

where the expectation is over the observation distribution induced by b and a . The operator is the robust dynamic program on beliefs; V_k is concave and piecewise linear in b whenever the one-step data are.

3. The support identity and exact sufficiency

The bridge to the set-valued theory runs through the support. Write $\text{supp}(b)$ for the states carrying positive mass, and let $\mathcal{W}_k$ denote the calculus's viable-set recursion: $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_k = \{B : \text{some admissible } a \text{ maps every } x \in B \text{ into } \mathcal{W}_{k-1}\text{-supported posteriors}\}$.

Theorem 3 (support identity). On finite models with deterministic kernels, for every action a and observation y ,

$$\text{supp}(b^+(\cdot | a, y)) = \text{Post}(\text{supp}(b), a, y),$$

the set-valued post-state of the calculus. Consequently $V_k(b) = 1$ if and only if $\text{supp}(b) \in \mathcal{W}_k$, while if $\text{supp}(b) \notin \mathcal{W}_k$ then every action loses a compatible state of positive mass, so

$$1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x).$$

Moreover $V_k(b)$ is non-increasing in k .

Corollary 4 (closed form on the delayed class). On the delayed hidden-regime instance — states (z, θ) with $\theta \in \{-1, +1\}$, prior $b_0 = (\frac{1}{2}, \frac{1}{2})$ from z_0 , no observation before T_{obs} , exact revelation at T_{obs} , blind-window holds $u \equiv \pm 1$, and unit drift against the floors — the belief support follows the two branches $z_0 \mp k$, and

$$V_k(b_0) = \begin{cases} \frac{1}{2} (\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1]), & k \leq T_{\text{obs}}, \\ \frac{1}{2} (1 + \mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]), & k > T_{\text{obs}}. \end{cases}$$

In particular $V_k(b_0) = 1$ exactly on the viable region $z_0 \geq 1 + T_{\text{obs}}$, recovering the deterministic verdict as the value-one level.

Derivation of Corollary 4. While $k \leq T_{\text{obs}}$ no observation arrives, so a policy commits to one blind hold $u \equiv \pm 1$: under $u \equiv +1$ the $\theta = +1$ branch sits at $z_0 + k$ at the horizon and the $\theta = -1$ branch at $z_0 - k$; under $u \equiv -1$ the roles swap. Each branch carries mass $\frac{1}{2}$ and survives exactly when it stays at or above the floor, giving

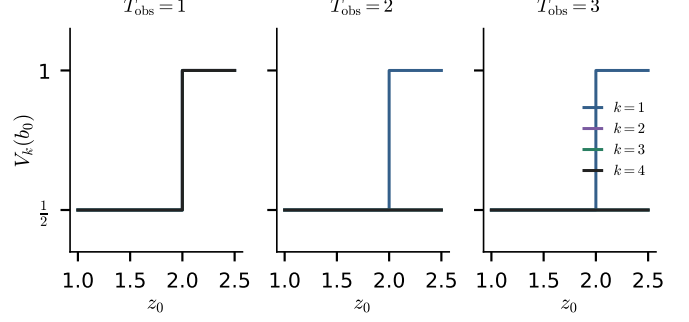


Figure 1: The closed-form values of Corollary 4 on the delayed grid: flat at $\frac{1}{2}$ while a branch falls, jumping by exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$, constant beyond.

$\frac{1}{2}(\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1])$, and the adversary cannot do better than pick the falling branch because the infimum over disturbance strategies selects the branch. For $k > T_{\text{obs}}$, revelation at T_{obs} splits the belief: the learned law $u = \theta$ regenerates the realized branch, so survival reduces to surviving the blind window of length T_{obs} on both branches — the mass- $\frac{1}{2}$ falling branch sits at $z_0 - T_{\text{obs}}$ at revelation, whence the indicator $\mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]$ weighting the certainly-saved mass $\frac{1}{2}$. $\square$

4. Exact piecewise-linear values

Proposition 5 (rational alpha-vectors). Over rational input data, the backward recursion of Theorem 2 propagates finite sets of rational alpha-vectors: each V_k is the maximum of finitely many rational affine functions of b , every value is an exact rational, and no floating point is required at any horizon. In the verification campaign the alpha-vectors of all audited instances — 384 vectors across the two-floor instance and the delayed grid — are exact rationals.

Remark 6 (dimension). On the two-floor instance the value function on the one-dimensional simplex is $\max(b_0, b_1)$ for all audited horizons (Figure 5); the witness alpha-vectors live in the two-coordinate display, with the suppressed bottom coordinate identically zero.

On the delayed grid the campaign evaluates $\alpha_k(z_0, T_{\text{obs}})$ at all $16 \times 3 \times 4 = 192$ cell-horizon combinations of $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$, $k \in \{1, 2, 3, 4\}$. The realized vectors take exactly three types: (1, 1) (both branches savable in the blind window), 72 occurrences; and the mirror pair (1, 0) and (0, 1) (one branch savable), 156 occurrences each — the symmetry between the two mirror types is exact, $72 + 156 + 156 = 384$. Table 1 carries the complete value table; Figure 1 displays the closed forms.

5. The deterministic limit

As the observation vanishes the value recursion is replaced by a distinct robust operator.

Proposition 7 (survived-mass formula). On the delayed class with blind window T_{win} , the k -step value of the prior equals the maximal survived mass over blind action sequences:

$$V_k(b_0) = \max_{u_{1:k}} \left[w \mathbb{K}[z_0 + \sum_{t \leq k} u_t \geq 1] \right]$$

Table 1: The complete audited value table on the delayed grid: the exact $V_k(b_0)$ (values 1 or $\frac{1}{2}$) for all 16 grid states, three blind-window lengths, four horizons — 192 cell-horizon combinations, each regenerated by script and each equal to both the alpha-vector recursion and the closed form of Corollary 4.

z_0	$T_{\text{obs}} = 1$				$T_{\text{obs}} = 2$				$T_{\text{obs}} = 3$			
	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$	$k=1$	$k=2$	$k=3$	$k=4$
1.0	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.2	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.3	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.4	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.5	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.6	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.7	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.8	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
1.9	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
2.0	1	1	1	1	1	1	1	1	1	1	1	1
2.1	1	1	1	1	1	1	1	1	1	1	1	1
2.2	1	1	1	1	1	1	1	1	1	1	1	1
2.3	1	1	1	1	1	1	1	1	1	1	1	1
2.4	1	1	1	1	1	1	1	1	1	1	1	1
2.5	1	1	1	1	1	1	1	1	1	1	1	1

Table 2: The asymmetric audit at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$, $b_0 = (\frac{3}{10}, \frac{7}{10})$: survived mass of each blind two-step sequence. Every sequence loses a branch; the value is $\frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass.

Sequence	Survived mass	Lost branch
(+1, +1)	$\frac{3}{10}$	falling branch at step 1
(+1, -1)	$\frac{3}{10}$	falling branch at step 1
(-1, +1)	$\frac{7}{10}$	falling branch at step 2
(-1, -1)	$\frac{7}{10}$	falling branch at step 2

$$+ (1 - w) \mathbb{I}[z_0 - \sum_{t \leq k} u_t \geq 1],$$

with $w = \frac{1}{2}$ on the symmetric prior and the supremum attained on pure sequences. This is a different operator from the robust verdict recursion of the deterministic theory, and it recovers that theory’s verdicts as its 0/1 specialization.

On the two-floor chance instance — states $\{0, 1\}$, actions $u \in \{\frac{2}{5}, \frac{3}{5}\}$, state 0 surviving exactly the lower action and state 1 exactly the higher — the one-step servable sets are $\Gamma_1 = \{(1, 0), (0, 1)\}$ and $V_k(b_0) = \frac{1}{2}$ for all audited $k \leq 5$: the chance deficit $\delta = \frac{1}{2}$ is attained, and the deficit equals the minimal mass (Figure 5).

The asymmetric audit prices the deficit beyond the symmetric case. At $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ with prior masses $w = \frac{3}{10}$ and $\frac{7}{10}$, the four blind two-step sequences survive $\{\frac{3}{10}, \frac{3}{10}, \frac{7}{10}, \frac{7}{10}\}$ (Table 2): every sequence loses a branch, so $V_2(b_0) = \frac{7}{10}$ and the deficit $\frac{3}{10}$ equals the minimal mass — the degenerate bound of Proposition 8 attained off the symmetric prior (Figure 2).

6. Certificates as deficit bounds

The obstruction certificates bound the deficit from below.

Proposition 8 (deficit lower bounds). *The following hold on the audited instances. (i) Chance-constrained common-action bound: if no single action keeps all compatible states safe, the deficit is at least the minimal compatible*

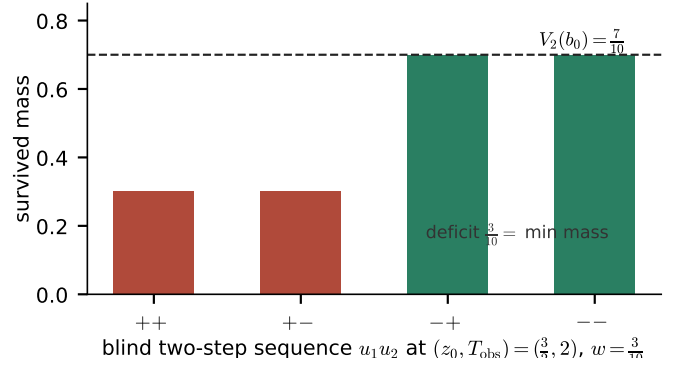


Figure 2: The asymmetric audit: survived mass by blind two-step sequence at $(\frac{3}{2}, 2)$ with $w = \frac{3}{10}$. The best sequences save the heavy branch only; the deficit $\frac{3}{10}$ is the minimal mass, attained.

mass. (ii) Degenerate minimal-mass bound: *in the deterministic limit the deficit is at least the minimal mass of the belief, attained whenever exactly one branch is savable (the two-floor instance attains $\delta = \frac{1}{2}$; on the delayed grid at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ every blind sequence loses a branch, so the deficit is at least the minimal mass $\frac{3}{10}$ of the audited asymmetric prior, and Table 2 shows the bound attained).* (iii) Monotone deficit: *the deficit is non-decreasing in the horizon on the audited class, and on the certified nonviability region of the delayed grid the deficit is constant.*

7. Agreement on the delayed-regime class

The class declaration is part of the model: the blind-window policy may be required to hold one action throughout the window (the declared hold class), or may act sequentially without observation, or may act with open-loop time-variation.

Theorem 9 (certificate-value agreement). *On all 48 audited cells (z_0, T_{obs}) with $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, and horizons $k \leq 4$: (i) the class-restricted alpha-vector recursion of Proposition 5 reproduces the closed form of Corollary 4; (ii) the certificate partition of the grid*

coincides with the value boundaries; (iii) the attaining witness and the value's jump loci trace the certificate boundary exactly, the jump being exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$ and the value constant on either side.

Derivation of the jump loci. Fix T_{obs} and k . On the closed form: if $k \leq T_{\text{obs}}$ the value is $\frac{1}{2}(\mathbb{K}[z_0 \geq 1+k] + 1)$, the second indicator being identically one on the grid; it jumps by exactly $\frac{1}{2}$ at $z_0 = 1+k$. If $k > T_{\text{obs}}$ the value is $\frac{1}{2}(1 + \mathbb{K}[z_0 \geq 1+T_{\text{obs}}])$, jumping by exactly $\frac{1}{2}$ at $z_0 = 1+T_{\text{obs}}$. Both loci are $z_0 = 1 + \min(k, T_{\text{obs}})$; between jumps both expressions are constant in z_0 . The witness family changes exactly at the same loci: the attaining one-step vector is $(1, 1)$ on $z_0 \geq 2$ and $(1, 0)$ below, and at horizon k the change sits at $z_0 = 1 + \min(k, T_{\text{obs}})$ — in particular at $z_0 = 3$, not 2, for $k = 2$ with $T_{\text{obs}} \geq 2$. $\square$

Theorem 10 (class declaration). *The declared hold class and the unrestricted sequential blind class coincide at the one-step horizon. For horizons $k \geq 2$ they differ exactly on the timing cells — those with $2 \leq z_0 < 1 + T_{\text{obs}}$ — and there open-loop time-variation within the blind window lifts the value to one: an alternating hold sequence keeps both branches at or above the floor whenever $z_0 \geq 2$, which no constant hold does. On the common-action cells $z_0 < 2$ the restriction costs nothing — no blind sequence saves both branches — and on the viable cells both classes attain one. On the audited grid the difference comprises exactly the 12 timing cells at each of $k = 2, 3, 4$: 36 cell-horizon combinations (Figure 3).*

Derivation of Theorem 10. At $k = 1$ both classes choose one blind action, so they coincide. Let $k \geq 2$ and write the two branch positions after a blind sequence $u_{1:\ell}$, $\ell = \min(k, T_{\text{obs}})$, as $z_0 \pm \sum_t u_t$. If $z_0 < 2$, some branch sits below $2 - (z_0 - 1)$ steps from its floor, and since the partial sums of any ± 1 -sequence move one branch down whenever they move the other up, one branch reaches its floor: no blind sequence saves both, both classes give $\frac{1}{2}$, no difference. If $z_0 \geq 1 + T_{\text{obs}}$ the constant hold already survives the window on both branches and both classes attain one. On the timing cells $2 \leq z_0 < 1 + T_{\text{obs}}$: a constant hold drives the falling branch to $z_0 - \ell < 1$ (deficit), while the alternating sequence has partial sums in $\{0, \pm 1\}$, so both branches stay at or above $z_0 - 1 \geq 1$ through the window, and revelation at T_{obs} then regenerates both: the unrestricted class attains one while the declared class does not. This locates the difference setwise. $\square$

8. Adaptive observation and hidden parameters

Observation is a design variable. On the declining instance the state carries a hidden parameter $\theta \in \{-1, +1\}$ gating the correctable component of a $-\frac{1}{10}$ drift, and the belief lives on the product (z, θ) ; the obstruction is the emptiness of the joint floor-servable set over (x, θ) cells.

Proposition 11 (probes and learning deadlines). *On the audited grid: with θ observed, every cell is viable (16 of 16). Under one-probe endogenous learning, the probe reveals θ exactly — the branch separation is 2, so one observation splits the belief with no statistics — but survival*

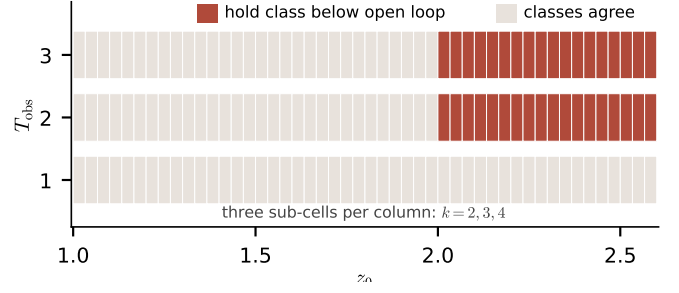


Figure 3: The class declaration on the audited grid: red cells mark cell-horizon combinations where the declared hold class and the unrestricted open-loop class differ — exactly the timing cells ($2 \leq z_0 < 1 + T_{\text{obs}}$) at horizons $k = 2, 3, 4$ (three sub-cells per column); 36 combinations in all. The classes agree at $k = 1$, on the common-action cells, and on the viable cells.

Table 3: The learning-deadline audit: the exact threshold $\frac{21}{10} + \frac{T_{\text{learn}}}{10}$ and the resulting kernel on the grid at each deadline.

T_{learn}	Threshold	$ K $	Kernel K on the grid
0	21/10	5	{2.1, 2.2, 2.3, 2.4, 2.5}
1	22/10	4	{2.2, 2.3, 2.4, 2.5}
2	23/10	3	{2.3, 2.4, 2.5}
3	24/10	2	{2.4, 2.5}
4	25/10	1	{2.5}

forces $z_0 \geq 21/10$. Under exogenous revelation at horizon T_{learn} , viability holds exactly when

$$z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}, \quad T_{\text{learn}} = 0, 1, 2, 3, 4,$$

the learning-deadline identity: late resolution acts as the timing obstruction's observation deadline, and the kernel shrinks exactly linearly and antitone in it, $|K| = 5, 4, 3, 2, 1$ on the grid for $T_{\text{learn}} = 0, 1, 2, 3, 4$ (Table 3 and Figure 4). A two-patch instance with unknown regeneration rate $g \in \{1, 2\}$ shows the same phenomenon on the audited two-patch belief pair: unknown-regeneration nonviable, (x, g) -observed viable.

9. Verification methods

Every value, bound, identity, threshold, figure, and tabulated entry in this paper is regenerated by deterministic scripts in exact integer and rational arithmetic, standard library only. Two per-system scripts cover the audited instances — fifteen check families on the stochastic values (the two-floor instance, piecewise-linearity, the deterministic limit, the 48-cell agreement, the partition coincidence, the witness family, the deficit identity, the unrestricted class, rationality of all 384 campaign alpha-vectors, deficit monotonicity, the oscillation identity, model completeness, the selector split, dimension consistency, and the jump loci) and six check families on the adaptive-observation instance (theta observed, the probe threshold, the learning-deadline identity, injective revelation, the antitone kernel chain, and the unknown-regeneration instance). A master script chains the two, then recomputes independently from the definitions: the closed-form grid values are reproduced by direct branch-mass enumeration for every cell with $k \leq T_{\text{obs}}$, the alpha-vector recursion is re-derived and matched against the

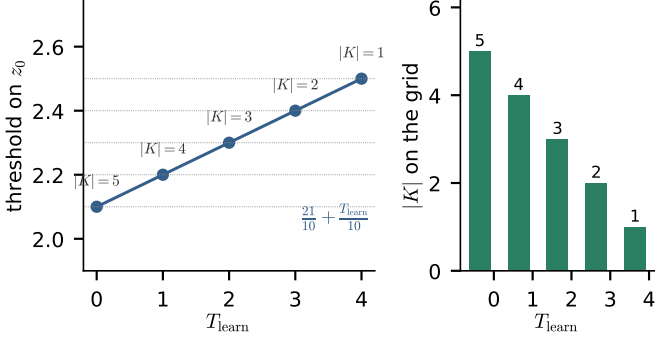


Figure 4: The learning-deadline identity: the viability threshold increases exactly linearly in T_{learn} (left), and the kernel on the grid shrinks linearly and antitone, $|K| = 5, 4, 3, 2, 1$ (right).

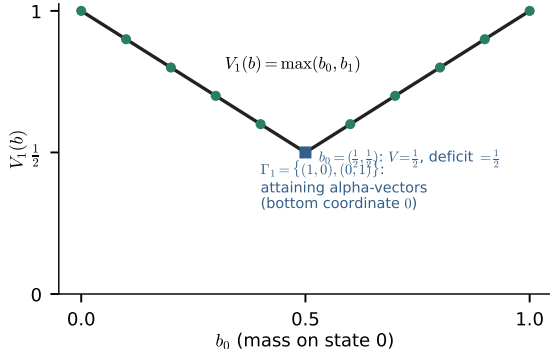


Figure 5: The two-floor chance instance: the one-step value is $\max(b_0, b_1)$, attained by the alpha-vectors of $\Gamma_1 = \{(1, 0), (0, 1)\}$ with zero bottom coordinate (green probes); at the symmetric prior the value is $\frac{1}{2}$ and the deficit equals the minimal mass.

closed form on all 192 audited combinations, the class-difference set is re-enumerated by brute force (36 combinations), the jump loci are recomputed from the closed form on the grid, the 384-vector type census (156, 156, 72) is re-derived, every entry of Tables 1, 2, and 3 is projected against the recomputation, and every headline number is asserted present in the text. A companion figure-and-table script recomputes and asserts each plotted or printed quantity from the same primitives before drawing; the two scripts agree on every shared quantity.

10. Conclusion

The belief-state safety value carries the obstruction calculus into the probabilistic setting without loss of exactness. The support identity ties the value-one level to the viable-set recursion; rational alpha-vectors keep every audited value exact, and the complete value table shows the two-valued structure the certificates predict; the certificates survive as deficit lower bounds, attained on natural instances on both the symmetric and the asymmetric prior; the audited delayed-regime class agrees with its closed form cell by cell, with the class declaration responsible for exactly the 36 timing combinations; and the adaptive-observation audit prices late resolution exactly — a linear learning deadline on the viable set, $|K| = 5, 4, 3, 2, 1$. The observation layer, not the forecast layer, sets the value; the design rules are the same ones the deterministic audits impose, now with a price attached and every number

in the tables open to script-level re-inspection.

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Declarations

Funding

None.

Competing interests

None.

Data availability

No external data were used.

Code availability

The verification script [paper2_belief_state_v2_verification.py](#) and the figure-and-table generation script [paper2_belief_state_figures.py](#) are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); they reproduce all values, bounds, thresholds, figures, and tabulated entries verbatim.

AI declaration

GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.